

1. Find the sum of all elements in an array.
2. Find the average of array elements.
3. Find the maximum/minimum element in an array.
4. Search for a given element (linear search).
5. Count the occurrences of a specific element.
6. Replace all occurrences of a specific element with another number.
7. Check if an array contains a given number.
8. Count how many elements are even and how many are odd.
9. Separate even and odd elements into two arrays.
10. Copy elements from one array to another.
11. Merge two arrays into a third array.
12. Compare two arrays for equality.
13. Find the index of the first occurrence of a given element.
14. Find the index of the last occurrence of a given element.
15. Shift all elements to the left by one position.
16. Shift all elements to the right by one position.
17. Sort an array in ascending order .
18. Sort an array in descending order.
19. Find the second largest element in an array.
20. Find the second smallest element in an array.
21. Remove duplicates from an array .
22. Count frequency of each element.
23. Reverse an array in place without using extra space.
24. Rotate an array by k positions to the right.
25. Rotate an array by k positions to the left.
26. Find the largest three elements in an array.
27. Find the smallest three elements in an array.
28. Find the sum of even and sum of odd elements separately.
29. Replace negative numbers with zero.
30. Replace all zeroes with ones.
31. Find the difference between the largest and smallest element.
32. Find all pairs of elements whose sum is equal to a given number.
33. Check if an array is sorted (ascending).
34. Count the number of elements greater than a given number.

35. Move all zeroes to the end without changing order of other elements.
36. Rearrange array so that positive and negative numbers alternate.
37. Find the majority element (element occurring more than $n/2$ times).
38. Find the intersection of two arrays.
39. Find the union of two arrays.
40. Find missing number in a sequence from 1 to n.
41. Find the duplicate number in an array (with one duplicate).
42. Find the element that appears only once while others appear twice.
43. Find the sub array with maximum sum (Kadane's algorithm).
44. Count the number of sub arrays with a given sum.
45. Find the length of the longest sub array with all distinct elements.